

B. REKÜRANS BAĞINTILARI KULLANARAK ÇÖZÜM

Aşağıdaki soruları rekürans bağıntıları kullanarak kuvvet serileri ile çözünüz.

$$1. \quad \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 + 2)y = 0$$

$$y = c_0 \left(1 - x^2 + \frac{1}{4}x^4 - \dots\right) + c_1 \left(x - \frac{1}{2}x^3 + \frac{3}{40}x^5 - \dots\right)$$

Çözüm:

$$y'' + x y' + (x^2 + 2)y = 0$$

$$y = \sum_{n=0}^{\infty} c_n x^n, \quad y' = \sum_{n=1}^{\infty} n c_n x^{n-1}; \quad y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + x \sum_{n=1}^{\infty} n c_n x^{n-1} + (x^2 + 2) \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n + \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+2} + \sum_{n=0}^{\infty} 2 c_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1) c_{n+2} + (2+n) c_n + c_{n-2}] x^n = 0$$

$$\begin{cases} 2c_2 + 2c_0 = 0 \Rightarrow c_2 = -c_0 \\ 6c_3 + 2c_1 = 0 \Rightarrow c_3 = -\frac{1}{2}c_1 \end{cases}$$

Yaptığın kısım doğru, hata yok. Devamı:

$$* c_2 = -c_0, \quad * c_3 = -\frac{c_1}{2}$$

$$\text{ve } \sum_{n=2}^{\infty} [(n+2)(n+1) c_{n+2} + (2+n) c_n + c_{n-2}] x^n = 0$$

$$(n+2)(n+1) c_{n+2} + (2+n) c_n + c_{n-2} = 0; \quad n \geq 2$$

$n=2$ ise

$$12c_4 + 4c_2 + c_0 = 0 \Rightarrow c_4 = \frac{1}{4}c_0$$

$n=3$ ise

$$20c_5 + 5c_3 + c_1 = 0 \Rightarrow c_5 = \frac{1}{20} \cdot \left(\frac{3}{2}c_1\right) = \frac{3}{40}c_1$$

$n=4$ ise

$$6 \cdot 5 c_6 + 6c_4 + c_2 = 0 \Rightarrow c_6 = \frac{1}{30} \left(-\frac{c_0}{2}\right) = -\frac{c_0}{60}$$

Sonuç olarak

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + \dots$$

$$y = c_0 + c_1 x - c_0 x^2 - \frac{c_1}{2} x^3 + \frac{c_0}{4} x^4 + \frac{3}{40} c_1 x^5 - \frac{c_0}{60} x^6 + \dots$$

elde edilir.